

Major Advanced Topics

26433101

Ongonwou Fred

**Phd Candidate
2343003**

April 18, 2025

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields

(Phys. Rev. Lett. 130, 113201 (2023))

10.1103/PhysRevLett.130.113201

Hong Bin Yao^{1,2}, Qi-Wen Qu³, Zhao-Han Zhang¹, Jia-Wei Wang³, Jian Gao^{3,4}, Chen-Xi Hu¹, Hui Li^{3,*}, Jian Wu^{3,4,5}, Feng He^{1,5,†}

¹Key Laboratory for Laser Plasmas (Ministry of Education) and School of Physics and Astronomy, Collaborative Innovation Center for IFSA (CISIFSA) Shanghai Jiao Tong University, Shanghai 200240, China

²Key Laboratory of New Energy and Materials Research of Xinjiang Education Department, Xinjiang Institute of Engineering, Urumqi 830091, China

³State Key Laboratory of Precision Spectroscopy, East China Normal University, Shanghai 200241, China

⁴Chongqing Key Laboratory of Precision Optics, Chongqing Institute of East China Normal University, Chongqing 401121, China

⁵CAS Center for Excellence in Ultra-intense Laser Science, Shanghai 201800, China

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Theory

$$i \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[-\frac{\nabla^2}{2} + V(r) - i\mathbf{A}(t) \cdot \nabla \right] \Psi(\mathbf{r}, t) \quad (1)$$

- 2 different systems are used, so $V(r)$ is one of the following

$$V(r) = -1/r \quad \text{for H} \quad (2)$$

$$V(r) = -(1 + 1.231e^{-0.662r} - 1.325re^{-1.236r} - 0.231e^{-0.48r})/r \quad \text{for He} \quad (3)$$

- The interacting field is given by

$$A(t) = f(t) \left[-\frac{E_{\text{SH}}}{\omega_{\text{SH}}} \sin(\omega_{\text{SH}}t) - \frac{E_{\text{FW}}}{\omega_{\text{FW}}} \sin(\omega_{\text{FW}}t + \phi) \right], \quad f(t) = \sin^2(\pi t/T_p), \quad T_p = 20 \frac{2\pi}{\omega_{\text{SH}}}. \quad (4)$$

- ϕ is the relative phase between the two fields of frequency ω_{FW} and ω_{SH}

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Numerical representation

- The wavefunction $\Psi(\mathbf{r}, t)$ is expanded in a basis of B-splines functions, i.e

$$\Psi(\mathbf{r}, t) = \sum_{n\ell m}^{\infty} c_{n\ell m}(t) \Phi_{n\ell m}(\mathbf{r}), \quad \Phi_{n\ell m}(\mathbf{r}) = R_{n\ell}(r) Y_{\ell m}(\hat{\mathbf{r}}), \quad R_{n\ell}(r) = \sum_{i=1}^{N_r} d_i^{(n\ell)} B_i(r) \quad (5)$$

- The size of the box is $r_{\max} = 1500$ a.u., the time step is $\Delta t = 0.008$ a.u..
- The total excitation probability is given by

$$P_{\text{excit}} = \sum_{\substack{n\ell m \\ n>2}} \left| \left\langle \Phi_{n\ell m}^{\text{bound}}(\mathbf{r}) | \Psi(\mathbf{r}, T_p) \right\rangle \right|^2. \quad (6)$$

- The total ionization probability is given by

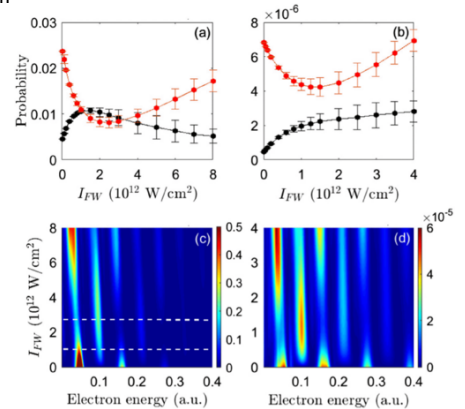
$$P_{\text{ioniz}} = 1 - \sum_{n\ell m} \left| \left\langle \Phi_{n\ell m}^{\text{bound}}(\mathbf{r}) | \Psi(\mathbf{r}, T_p) \right\rangle \right|^2. \quad (7)$$

- Convergence is achieved for a number of partial waves $\ell_{\max} = 40$ and $N_r = 1500$.

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Ionization and excitation probabilities

- Fig.1 upper panels show the ionization (red lines) and excitation (black lines) as a function of I_{FW} for H (a) and He (b)
 - Bullets represent the averaged probabilities over ϕ
 - Error bars denote the probability dispersion when ϕ varies
- Most of the ionization/excitation is caused by the SH
- ϕ changes the the probability but does not change the overall behaviour
- The reduction of the ionization yield can be seen as a function of I_{FW}



$$I_{SH} = 0.5 \times 10^{14} \text{ W/cm}^2 \quad \text{for H}$$

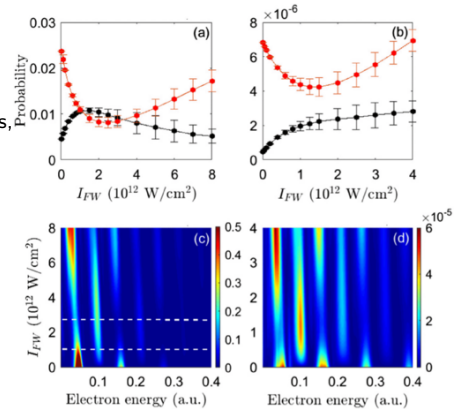
$$I_{SH} = 1.5 \times 10^{14} \text{ W/cm}^2 \quad \text{for He}$$

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Ionization and excitation probabilities

- Fig.1 lower panels show the I_{FW} -dependent photoelectron distribution (PED) for H (c) and He (d)
- The reduction of the yield can be seen as a function of I_{FW} ,
 - between 1 and 4×10^{12} W/cm² for the first peak in H
 - between 1 and 4×10^{12} W/cm² for the 1st and 2nd peaks, at different ranges for He

$$I_{SW} = 0.5 \times 10^{14} \text{ W/cm}^2$$



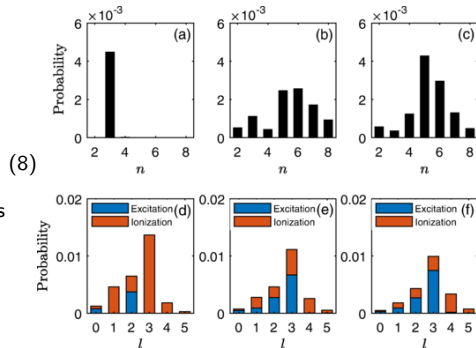
Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

n - and ℓ - resolved excitation probabilities for H

- Fig.2 n - and ℓ - resolved excitation probabilities for H
- 1st, 2nd and 3rd columns are for 3 different values of I_{FW} specified in Fig. 1(c)
- Those distributions are explained by the fact that

$$5\omega_{SW} + I_p > 0$$

- Distributions are explained by selection rules, propensity rules and random walking
- Introduction of I_{FW} enhances the overall excitation process
- Ionization reduction is shown by the lower panel



$$I_{FW}^1 = 0 \text{ W/cm}^2,$$

$$I_{FW}^2 = 10^{12} \text{ W/cm}^2,$$

$$I_{FW}^3 = 2.5 \times 10^{12} \text{ W/cm}^2$$

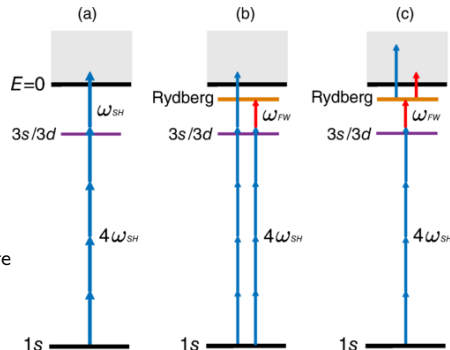
$$I_p = -0.5 \text{ a.u.},$$

$$\omega_{SW} = 0.114 \text{ a.u.}$$

Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Sketches of ionization for different I_{FW}

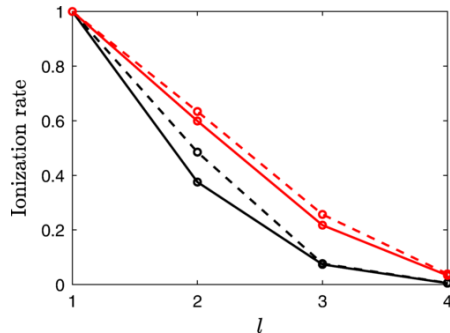
- (a) $I_{FW} = 0 \text{ W/cm}^2$. 5 photons absorbed for ionization with resonance at $n = 3$
- (b) $I_{FW} = 1 \times 10^{14} \text{ W/cm}^2$. $4\omega_{SH} + 1$ photons (or more) are absorbed for ionization at the Rydberg states.
- (c) $I_{FW} = 2.5 \times 10^{14} \text{ W/cm}^2$. $4\omega_{SH} + 2$ photons (or more) are absorbed for ionization at the Rydberg states. Increase of $P_{\ell=4}$



Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

The ℓ -dependent ionization rate of $n = 5$

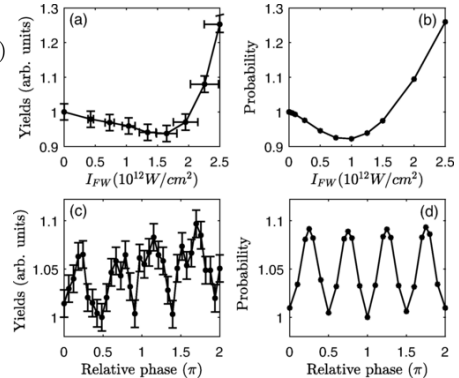
- Tug-of-war between excitation and ionization for $I_{\text{SH}} = 10^{13} \text{ W/cm}^2$ (black) and $I_{\text{FW}} = 10^{12} \text{ W/cm}^2$ (red)
- Ionization rate of Rydberg states $(5, \ell)$ for perturbation theory (solid lines) and TDSE (dashed lines)
- The greater the azimuthal quantum number of the initial state ℓ , the lesser is the ionization rate



Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Measured and simulated ionization probabilities of He

- Linearly polarized femtosecond laser pulse (35 fs, 800 nm, 1 kHz) delivered from a Ti : sapphire amplifier
- Fig. 5 shows the measured yield of He^+ from experiment (a) and simulation (b) as a function of I_{FW}
- Fig. 5 shows the measured yield of He^+ from experiment (c) and simulation (d) as a function of ϕ
 - Ionization (through $\ell=3$) may be done by 2 pathways
 - Their interference is thought to induce the fluctuation
- The overall ionization reduction is not caused by the phase but by the intensity



Multiphoton Ionization Reduction of Atoms in Two-Color Femtosecond Laser Fields (Phys. Rev. Lett. 130, 113201 (2023))

Conclusion

In this paper, the mechanism of ionization reduction in strong field is explored

- For atoms exposed to SH-FW laser fields, the weaker FW shunts the wavepacket to higher Rydberg states, though more loosely bound, are harder to ionize
- By tuning the intensity of the FW, the populations of the Rydberg and continuum states can be controlled
- So this is yet another study shedding light on ionization rate control.